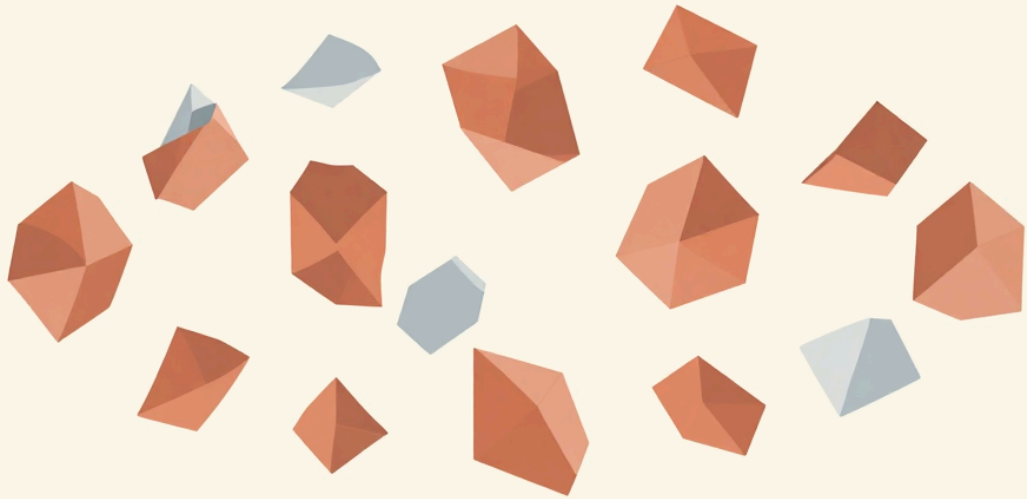




Attention is All You Need

PAPER EXPLAINED

Attention Is All You Need: No Recurrence Required

Attention Is All You Need

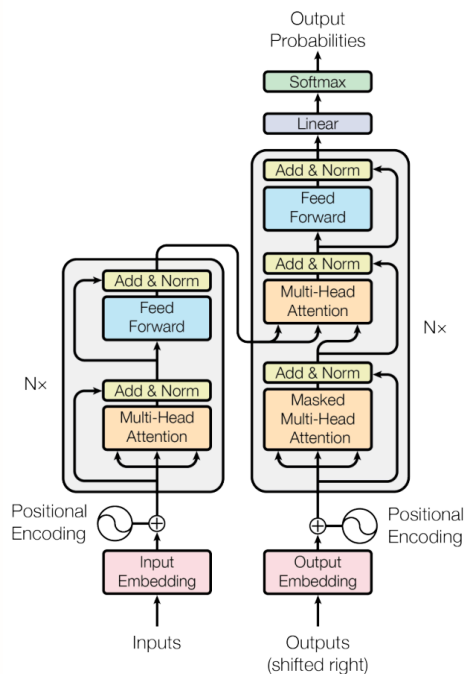
<https://arxiv.org/abs/1706.03762>

The Parallelization Bottleneck

- Recurrent models preclude parallelization due to their sequential nature.
- Transformers allow significantly more parallelization than recurrent models.

Introducing the Transformer Architecture

- The authors propose the Transformer, a model architecture that relies entirely on attention mechanisms.
- It eschews recurrence and is the first transduction model relying entirely on self-attention.

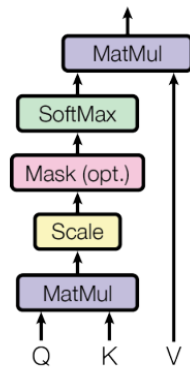


The Transformer - model architecture.

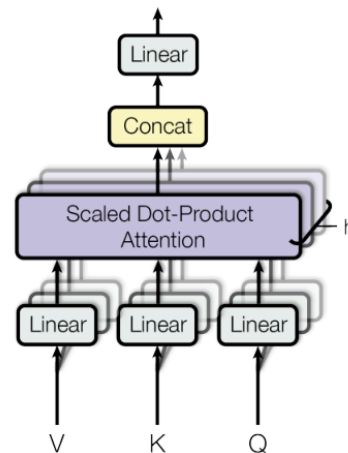
How Self-Attention Works

- Self-attention relates different positions within a single sequence.
- Scaled Dot-Product Attention uses dot products, scaling, and softmax to compute weights.
- Multi-head attention allows the model to attend to different representation subspaces simultaneously.

Scaled Dot-Product Attention



Multi-Head Attention



(left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

Transformer Translation Results

- Reaches state-of-the-art translation quality after twelve hours on eight P100 GPUs.
- Achieves better BLEU scores than previous models at lower training cost.

Big Transformer Results

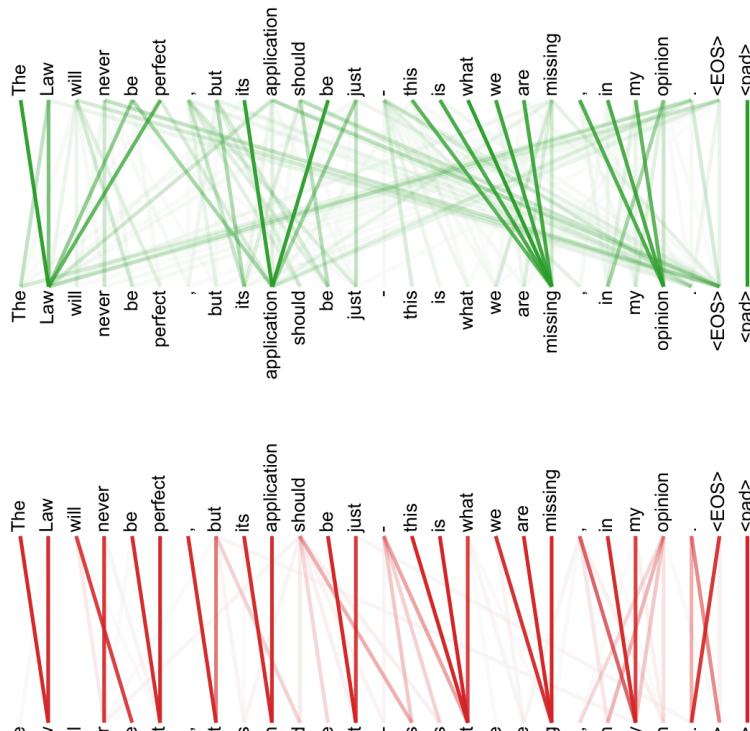
- The big Transformer achieves a BLEU score of 28.4 on WMT English-to-German.
- It scores 41.0 BLEU on English-to-French at less than one-quarter the training cost.

Limitations and Future Directions

- Reduced effective resolution occurs due to averaging attention-weighted positions.
- Future work extends the Transformer to other modalities and investigates local attention mechanisms.

Why This Matters for ML Engineers

- Self-attention enables greater parallelization than recurrent layers.
- Attention heads often learn syntactic and semantic structures.



Many of the attention heads exhibit behaviour that seems related to the structure of the sentence.